Predicting the infrared absorption of atmospheric gases

Introduction

In this activity you will estimate the bond lengths, bond angles, bond orders, and infrared spectra of seven atmospherically relevant gases: CO_2, N_2O, O_3, CH_4, O_2, N_2, and H_2O, and use the infrared spectral information to predict which molecules are greenhouse gasses.

When solar radiation strikes the earth some of the heat is transferred to the surface and warms it. Some of this solar radiation can be absorbed then re-emitted from the surface back to the atmosphere. This heat would then normally emanate out of the atmosphere and be lost to the earth forever. Instead, in the atmosphere, certain gases, called greenhouse gases, absorb the stray light and re-emit it which makes some solar radiation return to the surface a second time to reheat it. Earth, as a result, is warmer because of these greenhouse gases. This effect can be visualized the best from the spectra shown below. This is a radiative spectrum taken from space of the solar radiation coming from the earth's surface. If not for the greenhouse gases that are present in our atmosphere, the spectrum would be continuous and follow one of the dashed lines on the graph. These dashed lines represent the idealized radiative spectrum of the earth if it was giving off heat at different surface temperatures. It can instead be seen that there are massive dips from the baseline in three different regions around 580-750, 980-1100, and 1200-1500 wavenumbers. These dips correspond to certain missing wavelengths that are absorbed by these atmospheric gasses. All other wavelengths or radiation are not absorbed as effectively and are thus able to be detected from space as stray radiation. Your task in this lab is to model certain atmospheric gasses and predict which ones are most responsible for these regions.

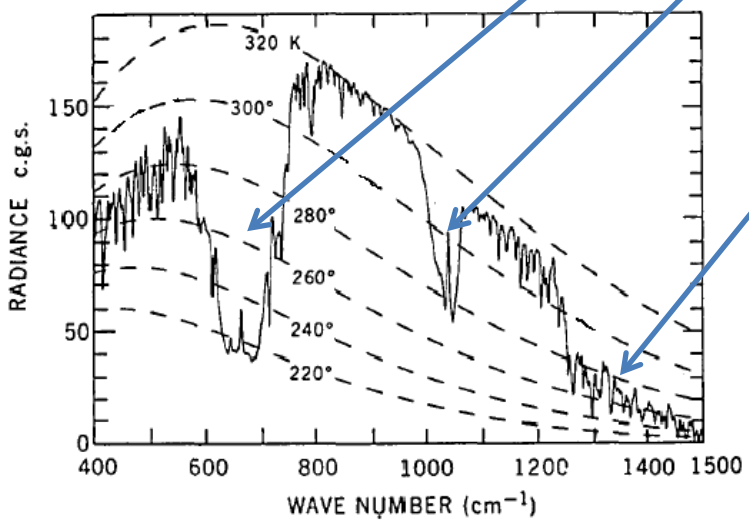


Figure 1. Earth infrared emission spectrum.

The IR spectrum shown in Figure 1 runs from low energy on the left to higher energy on the right, which is the opposite of modern convention and of the spectra you obtain in this experiment. The earth's emission is similar to a 320 K radiation source, but filtered by atmospheric gases. It is important to note that there is almost no energy at frequencies great than about 1200 cm^{-1}.

You will use the **Density Functional B3LYP method with a 6-31G* basis set**. Basis sets are used to make atomic orbitals and combined to create the molecular orbitals. Larger basis sets improve the results but require more computational resources. The method and basis set chosen in this experiment have

been selected to give good results with good time efficiency. Some approximations are better than others at determining certain structural properties with high accuracy and time efficiency; the B3LYP method is one of the better methods used to predict infrared spectra.

Infrared spectra

The absorptions in the IR are due to vibrations in the molecule. In a diatomic molecule like HCl, the only possible vibration is the stretch in which the bond length changes. This stretch occurs at a characteristic frequency that corresponds to the infrared region. But not all stretches absorb or emit IR radiation. In order for a stretch to be infrared-active, the stretch must change the dipole moment. The HCl stretch is IR-active because the H and Cl have different electronegatvities and HCl is a polar molecule. But the stretch of a homonuclear diatomic like H_2 does not change the dipole moment. In fact, the dipole moment of these molecules is zero.

Carbon dioxide is a linear molecule and at its equilibrium position, there is no dipole moment because the two bond dipoles are equal and opposite.

	O = C = O
Bond dipole	←+ +→

The symmetric stretch in which both oxygen atoms move in or out the same distance causes no net change in the dipole moment and the symmetric vibration is not IR-active

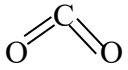
Equilibrium	O = C = O
Symmetric stretch	O == C == O
Symmetric stretch	O=C=O

But there is another stretch, called the asymmetric stretch, in which one oxygen atom move in and the other moves out.

Equilibrium	O = C = O
Asymmetric stretch	O=C == O
Asymmetric stretch	O == C=O

The asymmetric stretch is infrared active.

There is a third type of vibration, which is a bond bending, as opposed to a stretch.

Equilibrium	O = C = O
Bent	

If the bending motion results in a change of the dipole moment, then it will be IR-active. The CO_2 bend, sometimes referred to as a scissoring action, is IR-active.

In the following activity you will be asked to characterize vibrations as symmetric stretch,

asymmetric stretch or bend. Spartan will predict the intensity and if the intensity is zero, the vibration will not absorb IR radiation because it does not change dipole moment.

Table 1. Average composition of gases in the atmosphere.
From http://www.physicalgeography.net/fundamentals/7a.html

Gas Name	Chemical Formula	Percent Volume
Nitrogen	**N_2**	**78.08%**
Oxygen	O_2	**20.95%**
***Water**	H_2O	**0 to 4%**
Argon	Ar	**0.93%**
***Carbon Dioxide**	CO_2	**0.0360%**
Neon	Ne	**0.0018%**
Helium	He	**0.0005%**
***Methane**	CH_4	**0.00017%**
Hydrogen	H_2	**0.00005%**
***Nitrous Oxide**	N_2O	**0.00003%**
***Ozone**	O_3	**0.000004%**

*** Variable gases**

Recitation CHEM542 Name:

Consider gas phase N2, CO, O2 (singlet and triplet), N2O, CO2, H2O, O3, CH4

Compute their IR spectra at B3LYP/6-31G. *Use Data Table to record your results.*
Answer the following questions referring to Figure 1 and Table 1 as needed.

1. If nitrogen and carbon monoxide have the same number of electrons and the same bond order why is one IR active while the other is not?

2. Which of these atmospheric gases are greenhouse gases?

3. Which classification of vibrations is most likely responsible for the peak from 580-750 wavenumbers? Why might we not see these vibrations in diatomic molecules?

4. Which molecule is most responsible for the peak at 980-1100 wavenumbers? Take the experimental spectra into consideration here.

5. Which molecule is most responsible for the peak from 580-750 wavenumbers? Explain your answer.

6. Which molecules are responsible for the shallow drop off between 1200-1500 wavenumbers?

7. What conclusions, if any, can you make on the effect of bonding in infrared spectroscopy and the greenhouse effect?

Data Table

Experimental data has been included. The experimental harmonic frequencies include only IR-active frequencies.

Molecule	Bond length	Molecular angle	Harmonic frequencies	Calculated IR absorption intensity	[1]Harmonic vibration classification
Nitrogen					
	1.098 Å	N/A	N/A		
Carbon Monoxide					
	1.128 Å	N/A	2170 cm^{-1}		
Oxygen diamagnetic (singlet)	1.209 Å	N/A	N/A		
Oxygen paramagnetic (triplet)	1.207 Å	N/A	N/A		
Water					
	0.969 Å	104.478°	1595, 3657, 3756 cm^{-1}		
Methane					
	1.093 Å	109.471°	1534, 2917, 3019 cm^{-1}		
Carbon Dioxide					
	1.169 Å	180.000°	667, 1333, 2349 cm^{-1}		
Nitrous Oxide					
	N-N 1.128 Å N-O 1.184 Å	180.000°	589, 1285, 2224 cm^{-1}		
Ozone					
	1.264 Å	116.800°	705, 1042, 1110 cm^{-1}		

[1] Classify as asymmetric stretch, symmetric stretch, or bend.